

Meghapranay Raparla

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EDUCATION

Amrita Vishwa Vidyapeetham

2024 – 2028

B.Tech in Computer Science and Engineering · CGPA: 8.33/10

Coimbatore, India

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Computer Networks
- 145+ LeetCode problems solved — graphs, dynamic programming, recursion, trees

RESEARCH EXPERIENCE

Undergraduate Researcher

2025 – Present

Amrita Vishwa Vidyapeetham · Advisor: Dr. Harini N.

- Designed hybrid C2PA + CNN-ensemble media authentication pipeline; **94.2% detection accuracy** (F1=0.93) on 10,000+ adversarial deepfake samples
- Combined SHA-256 pixel-hash cryptographic verification with weighted voting over EfficientNet-B0, ResNet-50, and FFT-domain CNN; cryptographic bypass cuts avg. latency from 320 ms to ~160 ms
- Evaluated robustness under FGSM and PGD adversarial attacks ($\epsilon=0.01-0.03$); manuscript in preparation

Undergraduate Researcher

2025 – Present

Amrita Vishwa Vidyapeetham · Advisor: Dr. Amit Agarwal

- Fine-tuned YOLOv8 (COCO weights) on a curated 5,000+ image dataset for in-field strawberry disease detection; achieved **91.7% mAP@0.5** (precision 89.3%, recall 87.1%)
- Built IoU-based NMS pipeline (threshold 0.45) to eliminate duplicate detections on overlapping leaf regions

PROJECTS

IFDMS — Concurrent Focus Monitoring System

Python · asyncio · WebSockets · Flask · SQLite

- Built a layered concurrent architecture: asyncio WebSocket server on a background thread handles multiple simultaneous clients (Chrome extension + HUD); synchronous main loop runs 10+ ML pipeline modules with thread-safe shared state
- Implemented producer-consumer IPC: WebSocket callbacks write to shared *BrowserState*; main loop reads after each tick — no mutex contention on the hot path
- Real-time broadcast to multiple clients via *asyncio.gather*; blocklist re-synced on every client reconnect to prevent state divergence

C2PA AI Detector — Deepfake Authentication

Python · PyTorch · EfficientNet · OpenCV · Next.js

- Trained 4-model weighted ensemble (CNN3, CNN6, EfficientNet-B0, HybridFFT) on 10K+ real and synthetic video samples; 94.2% accuracy, F1=0.93 on adversarial test set
- HybridFFT model analyzes 2D FFT magnitude spectrum for GAN upsampling artifacts invisible to spatial models

MicroJava AST Interpreter

Java · JavaFX · Compiler Design

- Implemented full compiler pipeline: Lexer → Parser → AST construction → Tree visualizer (SVG) → Step-by-step interpreter with breakpoint debugger and symbol table / memory simulation

GeoDroneAI — Autonomous Drone Delivery Simulator

Python · A* Search · Geospatial

- Implemented A* pathfinding with weighted heuristics and dynamic obstacle re-planning for multi-drone fleet coordination; **34% faster** delivery times vs. naive shortest-path routing in simulation

SmartStore ERP — Enterprise Backend

Java · Spring Boot · JWT · JPA/Hibernate · MySQL

- RESTful ERP backend with JWT authentication, role-based access control, and Controller → Service → Repository layered architecture over Hibernate + MySQL; 7 normalized tables, 7 REST endpoints

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C

Concurrent Systems: asyncio, WebSockets, multi-threading, thread-safe shared state, producer-consumer IPC

AI / ML: PyTorch, TensorFlow, EfficientNet, YOLOv8, CNNs, ensemble learning, adversarial robustness

Systems & Backend: Spring Boot, Flask, REST API design, JWT/RBAC, MySQL, SQLite

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Compiler Design

Tools: Git, IntelliJ, VS Code, Linux, Agile/Scrum